

Security operation center fundamentals.

Security landscape :- Security operations (Secops) is a necessary function for protecting our digital way of life, for businesses and customers. Most organizations are responding with a fundamental shift to their cybersecurity approach - moving away from a collection of point solutions, ad hoc entities, and processes toward a more deliberate structure and the creation of dedicated Secops to manage and monitor a unified Security architecture.

What is security landscape?

=> Think of the security landscape like a battlefield map.

it includes

- organizations
- Technologies
- Internet + remote workers
- AI tools and automation
- Attackers.

But today's landscape is bit different.

- more cloud based
- more remote
- more connected
- more AI driven
- and more vulnerable.

What the Landscape Encompasses?

=> Risks

The biggest risk is a catastrophic breach, which may cause:

- Data theft (data exfiltration)
- Financial loss
- Reputation damage
- Loss of customers
- Legal penalties

* one major breach can destroy years of trust

=> Problems:

Organizations face problems like:-

- Too many security alerts
- Security tools not integrated
- Slow incident response
- Lack of skilled analytics

* without structure, security becomes chaotic.

=> Target objective

The main objective is:-

- Detect threats early
- Respond quickly
- Reduce damage
- protect sensitive data
- Maintain business continuity.

⇒ Deliverables

Security teams must deliver:

- Continuous monitoring
- Incident reports
- Threat analysis
- Risk assessments
- Improved security posture.

Security operations (Secops)

Security operations is the active defense team of an organization.

Their job is to:-

- Identify threats
- Investigate suspicious activity
- Mitigate attacks
- Continuously improve security.

⇒ Secops - Leads the charge

Secops professionals monitor:

- Networks
- Servers
- End point (laptops, desktops)
- Databases
- Applications
- Website
- cloud

Secops - Management and implementation:

Secops is a collaborative effort between security teams and operations teams that integrates tools, process, and technology for protecting our digital way of life.

it protects

- internal users
- partners
- customers
- systems
- Data

Security becomes a shared responsibility across the organization.

The ultimate goal:-

improve the organizations overall security posture.

Secops protects against security issues

1. Identify

Recognize suspicious alerts.

Example:- An unusual login attempt at 3AM.

action:- open an incident

2. investigate:-

Analyze logs, traffic, users behaviour

Questions asked:-

Is this real or false alarm?

Where did it start?

What systems are affected?

3. Mitigate

Stop the attack

Examples:-

- Block malicious IP
- Isolate infected systems
- Reset compromised passwords

4. continuously improve

After resolving the issue!

- update detection rules
- improve response time
- Learn from mistakes
- strengthn defense.

What is Security orchestration?

Security orchestration is the process of connecting different security tools together using automated workflows. Instead of tools working alone, they!

- communicate with each other
- share data
- follow automated response steps
- Reduce manual effort.

it helps security teams respond to incidents faster and smarter.

* Automating the process

Earlier:-

Secops engineer manually

- checks alerts
- Logs into 3-4 tools
- collects logs
- Analyze data
- Blocks IP.

This is waste of time

With automation

System automatically:-

- collects logs
 - checks threat intelligence
 - verifies infected machine
 - creates incident report
- in minutes.

* Reduces human workload

* Reduces alert fatigue

* Increase reponse speed

* improves accuracy.

* Important terminology:-

1. Security automation

using technology to perform security tasks automatically without human intervention.

Example:-

Auto block malicious IP

2. playbooks:-

Predefined step-by-step procedures for handling specific incidents.

Examples:-

if ransomware detected → isolate endpoint → alert admin → backup restore.

Playbooks ensure:-

- standardized response
- NO confusion
- faster handling

3. integration:-

how different tools connect and talk to each other

usually done using:-

- REST APIs
- webhooks
- connectors.

4) Ingestion

The process of collecting data from multiple sources into one platform

Example:-

Logs from firewall

Alerts from endpoint

cloud activity logs

Threat intelligence feeds

* Components & Technologies of security orchestration.

These components are backbone of orchestration systems.

1. SIEMC (Security Information and Event Manager)

it collects and analyzes logs from across the organization

it:

- Detects suspicious patterns

- generate alerts

- centralizes monitoring

* it's a brain that sees everything.

Threat Intelligence

it provides information about :-

- Malicious IP addresses
- Malware hashes
- phishing domains

• attack techniques

it helps answers :-

"Is this alert really dangerous?"

Endpoint security :-

protect devices like

- Laptops
- Desktops
- servers

it detects :

- malware
- suspicious behavior.

Network security :-

Monitors network traffic

it:

- Blocks malicious traffic
- Detects intrusion attempts
- Monitors unusual communication.

Security operations center Elements

A SOC is not just a room full of screens. It is a structured system built around six pillars that support the business.

* These pillars help divide responsibilities clearly so every stakeholder understands their role.

* The six pillars are:

1. Business
2. People
3. processes
4. interfaces
5. visibility
6. Technology

All of these connect back to one core idea:

The SOC exists to support the business. If the SOC does not reduce business risk, it is failing its purpose.

1. Business pillar (The foundation) -

This pillar defines:-

- why the soc exists
- how it is managed
- how success is measured

} are root elements.

it includes:

- Mission
- Governance
- planning
- Budget
- staffing
- facility
- metrics
- Reporting
- collaboration.

1. Mission - what are we doing?

The mission statement defines

- why the soc exists
- what it protects
- what results it must deliver.

it answers:-

- what actions will be taken?
- how will they be executed
- what value is provided to the business?

Example:

"detect, analyze, and respond to threats to protect company data and ensure business continuity"

2. Governance - How do we manage it?

Governance defines:

- policies
- standards
- compliance requirements
- oversight

it ensures:

- Rule are followed
- Responsibilities are clear
- Risk is managed properly.

3. Planning - How will we do it?

• planning defines:-

- strategy
- Road map
- Incident response plans
- Technology adoption

* It converts mission into action.

⇒ what is Metrics?

Metric (measuring what matters)

metrics shows whether the soc is effective.

* But not all metrics are good

* Poor Metrics (They drive wrong behavior)

1. Mean-Time-To-Respond (MTTR) in a network operations center (NOC) speed is everything

But in a SOC:-

- Rushing investigations reduces quality.
- Analysts may close incidents quickly without full analysis.

Speed $\neq$ security.

2. Number of incidents handled if analysts are ranked by volume:

- They may choose easy cases
- complex cases may be ignored

3. Number of firewall rules 10,000 firewall rules means nothing if:

Rule #1 says "allow Everything"

4. Number of SIEM feeds

more data $\neq$ better security,

if data isn't used, it's just noise

* Good metrics Build confidence

There are two types:

1. configuration confidence.

This means:

"Are our security controls properly configured?"

important questions:

* Are the security controls running?

Example:

A test port opened by a developer is accidentally left open.

That's risk.

* Are changes happening outside policy?
unauthorized changes reduce confidence.

* are tools configured to best practice?

security tools are not "install and forget"

They must be reviewed continuously.

* Are we ~~using~~ using the tool's features properly?

Many companies use only 30-40% of purchased features.

Example:-

70-80% of traffic is encrypted.

If SSL inspection is not used.

that traffic is invisible to analysts.

2. Operational confidence

which means

"Are our people and processes ready to handle a breach?"

* Events per Analyst Hours (EPAH)

Normal range: - 8-13 events/hour

if analyst handle 100/hour: -

- They are overwhelmed.
- Investigations are rushed.

This metric measures workload, not performance ranking.

* Are repeat incidents happening?

if the same alert appears again,

controllers are not being updated properly.

* Are known threats reaching SOC?

Known threats should be blocked automatically.

if SOC sees them,

prevention controls failed.

Executive (C&O) Questions

When a big vulnerability hits headlines

leaders want 'Simple answers' of

- Am I affected?
- What is the impact?
- What controls are protecting us?
- How long to fix it?

* Reporting

↳ Reporting proves value

it answers :-

"What did the SOC actually do?" *

There are :-

- Daily reports → operational activities
- Weekly reports → Trends and patterns
- monthly report → overall effectiveness.

* Simple summary

A SOC is built on six pillars, but everything ties back to business value.

GOOD SOCS:-

- Align with business mission
- Measures meaningful metrics
- Avoid vanity metrics
- Ensures tools are configured properly
- Ensures people are not overwhelmed
- provide clear reporting.